3.3.11 Amines (A-level only)

Amines are compounds based on ammonia where hydrogen atoms have been replaced by alkyl or aryl groups. This section includes their reactions as nucleophiles.

3.3.11.1 Preparation (A-level only)

Content	Opportunities for skills development
Primary aliphatic amines can be prepared by the reaction of ammonia with halogenoalkanes and by the reduction of nitriles. Aromatic amines, prepared by the reduction of nitro compounds, are used in the manufacture of dyes.	

3.3.11.2 Base properties (A-level only)

Content	Opportunities for skills development
Amines are weak bases. The difference in base strength between ammonia, primary aliphatic and primary aromatic amines. **Students should be able to:** • explain the difference in base strength in terms of the availability of the lone pair of electrons on the N atom.	

3.3.11.3 Nucleophilic properties (A-level only)

Content	Opportunities for skills development
Amines are nucleophiles. The nucleophilic substitution reactions of ammonia and amines with halogenoalkanes to form primary, secondary, tertiary amines and quaternary ammonium salts. The use of quaternary ammonium salts as cationic surfactants. The nucleophilic addition–elimination reactions of ammonia and primary amines with acyl chlorides and acid anhydrides. **Students should be able to** outline the mechanisms of: • these nucleophilic substitution reactions • the nucleophilic addition–elimination reactions of ammonia and primary amines with acyl chlorides.	